Agri-Doctor Pro: Multi-Crop Disease and Weed Diagnosis with Review Gates and an Independent Vision–Language Cross-Check

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***Abstract*—Crop-disease classifiers trained on curated leaf images often report near-perfect test accuracy, but a tool meant for farmers must also deal with photographs sent to the wrong crop model, confident labels that the image does not support, and field images unlike the training data. This paper presents Agri-Doctor Pro, a web application that identifies diseases and weeds in maize, rice and cotton. A crop router first selects one of three crop-specific TensorFlow Lite (TFLite) classifiers, and three decision rules are applied before any result reaches the user. Router suggestions for crops it is known to confuse require confirmation. The whole-leaf prediction is taken as the verdict, and four image quadrants are used only to check it. Finally, a vision–language model (VLM) examines the same photograph without access to the classifier output; the two readings are labelled as agreement, partial agreement, disagreement or uncertain, and a treatment plan is produced only when they are consistent. The cotton model was retrained as one ten-class EfficientNetV2-B0 network for five leaf conditions and five weed species. It achieved 95.57% accuracy (macro F1 0.959) on a held-out test set of 790 images, and 95.82% as a 7.0 MB TFLite file. The maize model achieved 97.28% on a validation split of 1580 images, or 94.98% when only maize-leaf images are counted. The rice model and the router have not yet been validated.**

***Index Terms*—convolutional neural networks, decision support systems, plant disease detection, transfer learning, vision–language models, weed identification**

I. Introduction

A classifier trained on laboratory leaf images will readily assign a confident label to a field photograph of a maize leaf with northern leaf blight. The confidence score alone says little about whether that label is correct. The image may have been sent to the wrong crop model, it may differ considerably from the training images, and softmax scores of modern networks are poorly calibrated [1]. Plant-disease models trained on curated collections are also known to lose much of their accuracy on images collected under other conditions [2], [3].

Agri-Doctor Pro started as our final-year project. The first version was a Streamlit dashboard that sent a leaf image to a crop-specific model, divided the image into quadrants, reported the most frequent disease or weed label among the quadrants, and asked a large language model for treatment advice [4]. When we tested it on photographs from outside the training sets, three problems appeared. The crop router assigned rice leaves to the maize model with 92–97% confidence. Quadrant crops of a healthy maize leaf were often classified as common rust, and since the old rule never classified the whole image, one or two such quadrants were enough to produce a disease diagnosis. The advice step then generated a treatment plan for whichever label it was given. In each case the user received a detailed answer with no indication that it might be wrong.

This paper describes the redesigned system and reports, for each component, whether it has been validated. The main contributions are as follows.

- A two-stage multi-crop pipeline in which each crop classifier predicts diseases and weeds together, with review gates that ask the user to confirm the crop when the router is unreliable or the prediction is weak.
- A decision rule in which the whole-image prediction is the verdict and quadrant predictions can confirm it but not replace it.
- A cross-check procedure in which a VLM receives only the image, its reading is compared with the classifier's by category rather than by averaging scores, and a prescription is generated only when the two are consistent.
- A retrained ten-class cotton model with 95.57% test accuracy, and an evaluation that keeps validated and unvalidated components separate.

II. Related Work

Leaf-image classification. Most leaf-disease classifiers fine-tune convolutional neural networks (CNNs) pretrained on ImageNet. Mohanty *et al.* [2] and Ferentinos [5] obtained accuracies above 99% on PlantVillage-type images, and Too *et al.* [6] compared several fine-tuned architectures on similar data. The same two early studies found that accuracy dropped sharply on images taken under different conditions. This led to field-image datasets such as PlantDoc [3] and to work on field imagery of maize northern leaf blight [7]. A broader survey of deep learning in agriculture is given by Kamilaris and Prenafeta-Boldú [8].

Multi-crop and weed recognition. Picon *et al.* [9] trained a single network on several crops and supplied the crop identity as an additional input, which gave better field accuracy than

separate per-crop models. Weed recognition has mostly been treated as a separate problem, for example on soybean field images [10], rangeland species [11] and weeds in cotton fields (CottonWeedID15) [12]. None of the disease studies cited above include weed classes.

Lesion- and patch-level analysis. Barbedo [13] classified individual lesions and spots instead of whole leaves and reported higher accuracy, as well as the ability to detect more than one disease on a leaf. This approach depends on each patch containing enough of the symptom. Our quadrant analysis is a simple form of patch analysis; Section III-C explains why it is now used only as a check.

Vision–language models. Multimodal models such as Gemini 2.5 [14] take an image as input and return a text description, so they can be asked to describe plant symptoms without task-specific training. Their accuracy on agricultural tasks is uneven. Arshad *et al.* [15] found that eight-shot prompting raised the best model's F1 score on plant-stress identification from 46.24% to 73.37%, and in AgroBench most open-source VLMs performed close to chance on weed identification [16]. For this reason we do not use the VLM as a classifier, only as a second opinion; when it disagrees with the CNN, the result deserves a closer look.

Reliability. Three known issues shaped our design: softmax outputs that are overconfident [1], networks that learn dataset-specific shortcuts [17], and test results inflated by leakage between training and test data [18].

III. System Design

Fig. 1 shows the pipeline. Its model-side logic is kept in a separate module with no Streamlit dependency, so it can be unit-tested with mock interpreters and regression-tested with the actual TFLite files [4].

A. Crop Routing and the Review Gate

A three-class router (cotton, maize, rice) proposes the crop model. The proposal is accepted automatically only when its confidence is at least 75% *and* the crop is not one that the router is known to confuse. On our regression photographs the router sent three of five rice leaves from Wikimedia Commons to the maize model, with confidences of 92–97%. For this reason, maize and rice proposals are always marked "needs review," and the crop selector starts empty so that the maize model cannot be run on a rice leaf with a single click. Cotton proposals above the threshold are accepted. The interface also records whether the crop was accepted, confirmed or set manually.

B. Crop Models and Input Contract

Each crop has a single classifier that predicts both leaf conditions and weeds. The maize model has six classes (blight, common rust, gray leaf spot, healthy, broadleaf weed and grass weed), the rice model eight and the cotton model ten (Section IV). All models accept raw 0–255 RGB images of 224×224 pixels and perform rescaling inside the network, which keeps preprocessing identical in training and deployment. A downloaded model file is used only if it carries the TFLite identifier, is larger than 100 kB and has one output per class label, and every prediction must sum to one within 0.03. A file that fails any of these checks is discarded and the existing model is kept.

C. Whole-Leaf Verdict and Quadrant Corroboration

In the first version, the image was split into four quadrants and the system reported the most frequent non-healthy quadrant label with a score above 50%. "Healthy" was reported only when no quadrant showed a problem, and the whole image was never classified. This rule failed repeatedly on the regression photographs. A quadrant crop loses the outline and venation of the leaf, and the maize model tended to classify such crops as common rust. One healthy maize leaf, for example, was classified as healthy from the whole image but as common rust in two of its four quadrants, and a leaf with northern leaf blight showed the same tendency.

In the current design the whole-image prediction is the verdict and the quadrants serve only as a check. Let $\hat{y}_w$ and p_w be the whole-image label and its softmax score, and $\hat{y}_q$ the label of quadrant q. The agreement count is

$$a = \sum_{q=1}^{4} \mathbf{1}\left[\hat{y}_q = \hat{y}_w\right], \tag{1}$$

and the result is marked for review when

$$p_w < \tau \quad \text{or} \quad a < a_{\min}, \tag{2}$$

where $\tau = 0.5$ and $a_{\min} = 2$. When $p_w < \tau$, the verdict is reported as "uncertain" instead of a disease or weed. Labels that occur only in quadrants, with a score above τ, are shown separately as possible secondary findings and do not affect the verdict. With this rule, all four quadrants of the common-rust photograph agree with the whole image and the result is accepted, whereas the blight photograph retains its correct whole-leaf label and is marked for review.

D. Field Context

The user can locate the field by searching for a place, clicking on a map or, on request, sharing the browser's location, and can draw the field boundary by hand. The Open-Meteo forecast API [19] supplies the current temperature, humidity, precipitation and sky condition, along with modelled volumetric soil moisture at a depth of 3–9 cm. Soil moisture is grouped into four bands (dry below 0.15, moist below 0.30, wet below 0.40 and waterlogged otherwise, in m^3/m^3). Since this value is a model estimate rather than a measurement, the interface says so, and the user may replace it with their own observation.

E. Independent VLM Cross-Check

At most two calls are made to Gemini (default model `gemini-2.5-flash`). The first call, at temperature 0.2, sends only the photograph, resized to at most 1024 px, together with a prompt that asks for an independent assessment. The

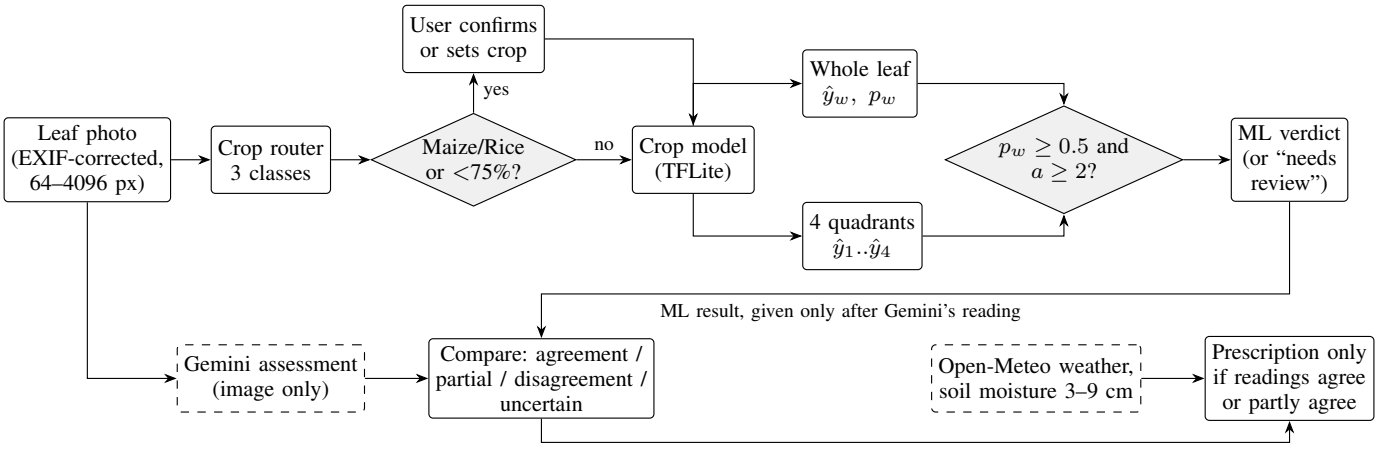


Fig. 1. Agri-Doctor Pro pipeline. Diamonds are review gates; dashed boxes are external services. The VLM never receives the classifier's output, and the ML verdict is never modified by the comparison.

classifier's result is intentionally not included. A JSON schema restricts the reply to a crop name, the closest label from the combined label sets of the three crops (or "other"/"unknown"), observed symptoms, supporting features, alternative explanations, image-quality problems, a flag for multiple conditions and a self-reported certainty level. Values outside the schema are replaced with "unknown" or "low."

The two readings are then compared and assigned to one of four categories:

- *Uncertain*: the ML verdict is below τ, Gemini reports low certainty, or Gemini cannot name a condition.
- *Disagreement*: Gemini sees a different crop or another plant; one reading is healthy and the other is not; or one finds a disease and the other a weed.
- *Agreement*: both give the same label, or both read the leaf as healthy (downgraded to partial if Gemini suspects several conditions).
- *Partial agreement*: both find a problem of the same kind but name it differently.

Confidence scores of the two systems are not averaged, and the interface describes agreement as supporting the result, not confirming it. The second call, which drafts a management plan from the weather, soil condition and both readings, is made only for agreement or partial agreement and only when the ML verdict is a disease or a weed. In the case of disagreement or uncertainty, the user is advised to take a closer, well-lit photograph or to consult an expert. If the API key is missing, the quota is exhausted or the network fails, the ML diagnosis is still shown along with an explanation. Successful responses are cached using the image hash and model name.

IV. Experimental Setup

A. Datasets

The datasets are listed in Table I. For cotton, we used the *Original Dataset* folder of a Kaggle cotton-disease collection [20], leaving out its augmented copies, together with the five largest classes of CottonWeedID15 [12]. We excluded the leaf variegation class because it contained only 116 images. The resulting 5264 images were divided into training, validation and test sets in a 70/15/15 ratio, stratified by class, *before* augmentation, and we checked that no file occurred in more than one set. The maize data consist of a Kaggle corn-leaf collection compiled from PlantVillage and PlantDoc images [21], 200 broadleaf-weed images and the grass class of a soybean weed dataset [10], with a fixed 20% kept as the validation split. The source of the broadleaf images is not recorded in the repository.

TABLE I
Datasets Used for the Crop Models

	Cotton	Maize	Rice
Classes (disease + weed)	5 + 5	4 + 2	6 + 2
Images	5264	7908	–
Train / val / test	3684 / 790 / 790	6328 / 1580 / –	–
Split	stratified 70/15/15	fixed 80/20	n/a
Sources	[20], [12]	[21], [10]	[22], [10]†

†Sources of the repository notebook runs; the deployed rice model's training data are not recorded.

B. Training

Both evaluated models use an EfficientNetV2-B0 backbone [23] pretrained on ImageNet, with all layers except the last 20 frozen. The backbone is followed by global average pooling, batch normalization, a 256-unit ReLU layer, dropout with a rate of 0.3 and a softmax layer. In the cotton model the batch-normalization layers were also kept frozen, leaving 1.14 M of the 6.25 M parameters trainable. It was trained with the Adam optimizer at a learning rate of 10^{-4}, a batch size of 32 and inverse-frequency class weights. Augmentation was applied only to training images and consisted of random flips, rotation (factor 0.1, i.e., up to $\pm 36°$), zoom (0.15) and contrast (0.15). Training was limited to 20 epochs with early stopping on validation loss (patience 4), and we kept the checkpoint

with the lowest validation loss, from epoch 19 (validation accuracy 95.06%).

The maize model was trained with Adam at a learning rate of 10^{-3} and heavier geometric augmentation (rotation, shift, shear, zoom and flip) for a fixed 15 epochs. Its 20% split was monitored as validation data during training, but it was not used for checkpoint selection or early stopping, and the weights from the last epoch were saved. Because the same split was also used in earlier training runs on different versions of the data, we treat it as a validation split and not as an untouched test set. Training was done on Google Colab, with a T4 GPU for the cotton model (about 3 min per epoch). For deployment, the cotton model was converted to TFLite with default dynamic-range quantization, while the maize model was converted without quantization (24.7 MB).

C. Unvalidated Components

Two deployed components, the rice model and the router, are not evaluated here. No training code for the deployed rice model exists in the repository; the rice notebook that is available trains custom three-block CNNs, which reached 61–71% test accuracy. In the regression tests, the deployed rice model classified only one of four labelled rice-disease photographs correctly, so all rice results are shown to the user as preliminary. For the router, the only evidence is the set of regression photographs mentioned earlier.

D. Metrics

We report accuracy, per-class precision, recall and F1 score, and their macro averages. All metrics were recomputed from the confusion-matrix counts printed or plotted in the training notebooks and matched the values reported there.

V. Results

A. Cotton: Ten Classes in One Model

On the 790 test images the cotton model achieved 95.57% accuracy, with macro precision, recall and F1 of 0.956, 0.963 and 0.959 (Table II). After conversion, the 7.0 MB TFLite model scored 95.82% on the same images. The difference of 0.25 percentage points corresponds to two images, and we do not interpret it as an improvement due to quantization.

The confusion matrix in Fig. 2 shows that all 35 errors occur within a group: leaf conditions were confused only with other leaf conditions, and weeds only with other weeds. In the leaf group, five of the six errors involve bacterial blight and leaf reddening, and accuracy on leaf images is 97.77%. Accuracy on weed images is lower, at 94.43%. The most frequent weed error is carpetweed predicted as waterhemp (7 of 115 images). These seven images make up more than half of waterhemp's 13 false positives and explain most of its low precision of 0.833.

Our earlier version used a six-class MobileNetV2 [24] model for cotton diseases, with a reported accuracy of 90.82% on 196 test images, and a separate five-class weed model with 82.98%. Since the label sets and test sets are different, these figures cannot be compared directly with ours. In practice, the main gain is that a single network now handles both tasks, and the user no longer has to state whether a cotton photograph shows a disease or a weed.

TABLE II
Cotton Model: Per-Class Test Results ($n = 790$)

	Class	P	R	F1	n
Leaf	Bacterial blight	0.921	0.946	0.933	37
	Curl virus	1.000	0.984	0.992	64
	Healthy	1.000	1.000	1.000	39
	Herbicide damage	1.000	1.000	1.000	42
	Leaf reddening	0.966	0.966	0.966	87
Weed	Carpetweed	0.990	0.896	0.941	115
	Morningglory	0.975	0.946	0.961	168
	Palmer amaranth	0.935	0.971	0.952	103
	Purslane	0.942	0.956	0.949	68
	Waterhemp	0.833	0.970	0.897	67
	Macro average	0.956	0.963	0.959	790

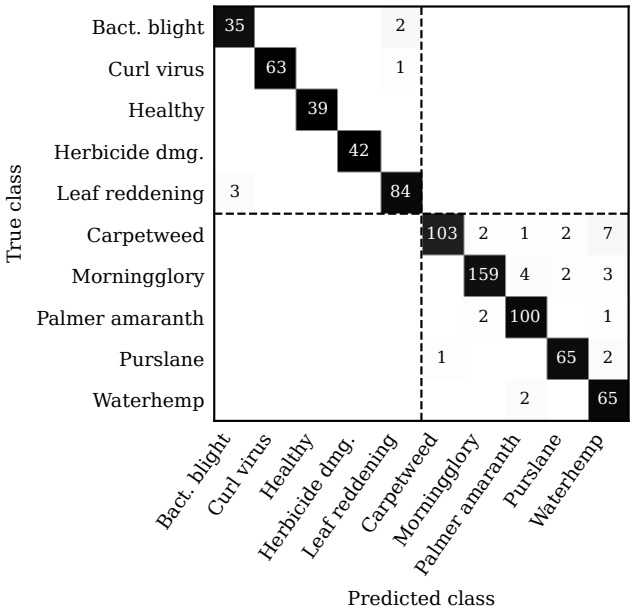


Fig. 2. Cotton confusion matrix on the test split (counts; shading is row-normalized recall). Dashed lines separate leaf conditions from weeds; no image was misclassified across this boundary.

B. Maize

The maize model achieved 97.28% accuracy on the validation split of 1580 images (macro F1 0.954), and the notebook reports a one-vs-rest AUC of 0.9979. This overall figure should be read with some caution. Grass-weed images, taken from a different dataset with a different camera setup, form 44.6% of the split, and every one of them was classified correctly. If only the 836 maize-leaf images are considered, accuracy is 94.98%. Most of the errors are between the two lesion diseases: 20 of the 114 gray-leaf-spot images were predicted as blight, giving gray leaf spot the lowest recall among the maize classes (0.798; Fig. 3). Healthy leaves were recognized with a precision and recall of 0.996. This is important in practice,

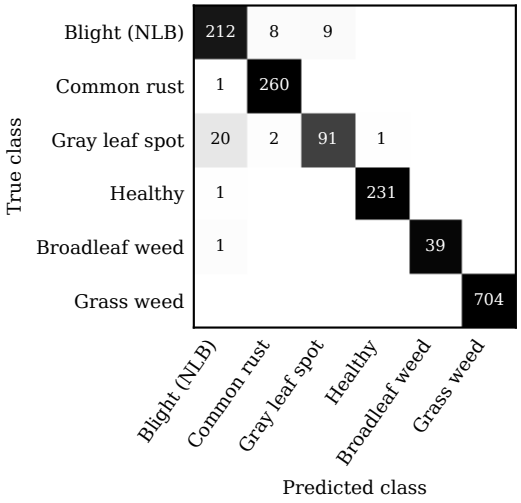


Fig. 3. Maize confusion matrix on the 20% validation split (counts; shading is row-normalized recall).

since a healthy leaf misclassified as diseased could lead to unnecessary spraying.

C. Behavior on Out-of-Distribution Photographs

Twelve Wikimedia Commons photographs of the three crops are used by the regression tests in the repository. This is enough to detect regressions but too few to estimate accuracy, so we report these as observations only. Both cotton photographs were sent to the cotton model without review. The maize model correctly classified the common-rust, blight and healthy maize photographs from the whole image. As noted in Section III-C, the rust photograph was accepted with all four quadrants in agreement, and the blight photograph kept its correct whole-leaf label but was marked for review. Under the old rule, the misclassified quadrants of the blight photograph would have produced a quadrant-level disease diagnosis.

VI. Discussion

Leaf versus weed separation. Fig. 2 shows no confusion at all between cotton leaf conditions and weeds. We are cautious about this result. Leaf and weed images come from different datasets, with different backgrounds, framing and cameras, and a network could separate the two groups using these cues alone [17]. The result therefore does not show that the model can tell a diseased cotton leaf from a weed in the same field photograph. The same applies to the perfect score on maize grass weeds.

Data splits. The cotton split was stratified, made before augmentation and checked for duplicate files, which addresses the most common cause of leakage [18]. It would not, however, catch near-duplicate photographs of the same plant in different sets, and public weed collections may contain such images. No duplicate check was run on the maize split, and its PlantVillage-derived images may include near-duplicates. A perceptual-hash check would address both cases. A check of this kind is already part of the rice training script in the repository, but it has not yet been run on real data.

Confidence and agreement. The 50% threshold in (2) is applied to uncalibrated softmax scores [1] and removes only the least confident predictions. A second, independent reading comes from the VLM, but we have not yet measured how often it agrees with the classifier or whether agreement is associated with correct diagnoses. Published benchmarks [15], [16] suggest that it will be least reliable for weeds. Until such measurements are available, the agreement categories should be taken as a prompt for closer inspection and not as evidence of accuracy.

Limitations. All reported accuracies were obtained on curated public datasets, not on field photographs collected for this study. The rice model and the router remain unvalidated. Field boundaries must be drawn by the user, soil moisture is a model estimate, and saved fields are kept only for the browser session. Inference time on mobile or edge devices has not been measured.

VII. Conclusion

This paper presented Agri-Doctor Pro, which identifies diseases and weeds in maize, rice and cotton using a crop router, crop-specific TFLite classifiers and an independent VLM reading. Review gates were added at the points where testing showed the earlier version to be unreliable. The retrained ten-class cotton model achieved 95.57% test accuracy (95.82% as a 7.0 MB TFLite model), and the maize model achieved 94.98% on the maize-leaf images of its validation split. In our experience, the decision rules mattered as much as the models. The system now asks the user to confirm a crop that the router may have chosen wrongly, does not allow quadrant predictions to override the whole-leaf prediction, and generates a treatment plan only when both readings support it. As future work, we plan to collect labelled field photographs of all three crops in order to evaluate the router, the rice model and the VLM agreement rate, and to retrain the router with additional rice images.

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